

Original URL: <https://www.kimi.com/chat/19af7869-8942-8f3b-8000-09c3316fb3b3>

[USER]

You're an experienced machine learning specialist who knows a lot about transformers, specifically BERT. Here is a jupyter notebook with some code to visualize BERT, and then answer some questions. Your task is to guide me through the complete process of answering the questions on the notebook, including what to run and what to give you.

We will be going through this step by step. For now, identify the questions I need to answer inside, and give me a plan for how we're going to go through this together.

Here, I gave it a pdf printout of the notebook.

[ASSISTANT]

Sure! Let's break down the process step-by-step. First, let's identify the questions you need to answer in the notebook:

Questions to Answer:

1. What similarities and differences do you notice in the visualizations between the examples in part (a)? Explore the queries, keys, and values to identify any interesting patterns associated with the attention mechanism.
2. How does attention differ between the different layers of the GPT model? Do you notice that the tokens are attending to different tokens as we go through the layers of the network?
3. Look at different layers of the BERT model in the visualizations of part (b) and identify different patterns associated with the attention mechanism. Explore the queries, keys, and values to further inform your answer. For instance, do you notice that any particular type of tokens are attended to at a given timestep?
4. Do you spot any differences between how attention works in GPT vs. BERT? Think about how the model architectures are different.
5. Look through the different layers of the two BERT networks above associated with sentence a and sentence b, and take a look at the queries, keys, and values associated with the different tokens. Do you notice any differences in the embeddings learned for the two sentences that are essentially identical in structure but different in meaning?
6. Do you notice BERT's bidirectionality in play?
7. Do you think pre-training the BERT helped it learn better representations?

A correct transcription of the questions in the notebook.

8. Do you notice different features being learned throughout the different attention heads of BERT? Why do you think this might be?
9. Can you identify any of the different features that the different attention heads are focusing on?
10. What differences do you notice in the attention patterns between the randomly initialized and trained BERT models?
11. Run the final cell in the notebook. What are some words or tokens that you would expect strong attention between? What might you guess about the gradients of this attention head for those words?

Plan to Answer the Questions:

1. Set Up Your Environment:

- Ensure you have the necessary libraries installed (bertviz, transformers, torch, etc.).
- Open the Jupyter notebook and run the initial cells to set up the environment and load the necessary models and libraries.

2. Run the Code for Visualizations:

- Run the cells that generate visualizations for GPT-2 and BERT models. This includes:
 - Simple sentences for GPT-2.
 - More complex sentences for GPT-2.
 - Sentence pairs for BERT.
 - Sentence pairs with reversed order for BERT.
 - Multiple attention heads for BERT.
 - Randomly initialized BERT model.

3. Analyze the Visualizations:

- For each question, carefully observe the visualizations generated by the code.
- Use the BertViz tool to hover over tokens, click on the + icon to see queries, keys, and values.
- Take notes on your observations, especially focusing on the attention patterns, differences between layers, and how attention changes with different inputs.

4. Answer the Questions:

The steps that it gave for me seem reasonable.

My intention is for it to tell me exactly what to do step by step and for me to then provide the visualizations, but I think “what to run and what to give you” might not be specific enough in my prompt, so it didn’t capture exactly what I wanted to do.

- Based on your observations, answer each question in detail.
- Provide specific examples from the visualizations to support your answers.

5. Document Your Findings:

- Write up your answers in a structured format, possibly in a separate document or within the Jupyter notebook itself.
- Include screenshots or descriptions of the visualizations to make your answers more comprehensive.

Step-by-Step Execution:

1. Run the Initial Setup Cells:

- Run the cells that install and import the necessary libraries (`bertviz`, `transformers`, `torch`, etc.).
- Load the pre-trained models (GPT-2 and BERT) and set up the visualization functions.

2. Generate Visualizations for GPT-2:

- Run the cells that visualize attention for simple and complex sentences.
- Observe the attention patterns and note any differences.

3. Generate Visualizations for BERT:

- Run the cells that visualize attention for sentence pairs.
- Observe how attention changes with different layers and heads.
- Experiment with reversing sentence order and observe the bidirectionality of BERT.

4. Analyze the Attention Mechanism:

- Use the BertViz tool to explore queries, keys, and values.
- Compare attention patterns between different layers and heads.
- Note any differences in attention for sentences with similar structure but different meanings.

5. Run the Randomly Initialized BERT Model:

- Run the final cell to reinitialize the BERT model weights.
- Compare the attention patterns with the trained model.

6. Answer the Questions:

- Use your observations to answer each question.

Seems to have understood the different subparts of the notebook (steps 2, 3, 4, and 5 roughly translate to parts 1, b, c, and d).

- Provide detailed explanations and examples from the visualizations.

Next Steps:

- **Run the Initial Setup Cells:** Start by running the cells that install and import the necessary libraries and load the pre-trained models.
- **Generate Visualizations for GPT-2:** Move on to running the cells that generate visualizations for GPT-2 and start observing the attention patterns.

Let me know when you're ready to proceed with running the initial setup cells!

[USER]

To clarify, I will be your “tool” for you to become an agent. I will provide you what you can visualize for each section/part, you tell me a couple of things you want to visualize (from the code), and I will give you the results, you will interpret the results, and potentially ask me for more results later. After doing so, you will answer the questions for that part.

Let's start with part a, which has the following questions: 1. What similarities and differences do you notice in the visualizations between the examples in part (a)? Explore the queries, keys, and values to identify any interesting patterns associated with the attention mechanism. 2. How does attention differ between the different layers of the GPT model? Do you notice that the tokens are attending to different tokens as we go through the layers of the network?

Here are the screenshots of the three cells of that subpart. Layer and head each range from 0 to 11, inclusive. Let me know which combinations you want to visualize for each sentence, and I will get them to you. Outside of the colors, I can also do the following operations for you: ” BertViz Usage (from their github page): - Hover over any of the tokens on the left side of the visualization to see what tokens are being paid attention to at that moment. - Then, click on the + icon that is revealed when hovering. This will reveal the query vectors, key vectors, and intermediate computations for the attention weights (blue=positive, orange=negative). - Once in the expanded view, hover over any other token on the left to see the associated attention computations. - Click on the Layer or Head drop-downs to change the model layer or head (zero-indexed). ”

Please give me some things to visualize, but please try to keep the amount of screenshots you ask for relatively minimal (say, 5 to 10 things to visualize) so I don't have to give too many screenshots.

[ASSISTANT]

Great! Let's start by visualizing some specific combinations of layers and heads for each sentence to analyze the attention patterns. We'll focus on a few key layers and heads to keep the number of visualizations manageable.

Here, I clarify that my goal is for Kimi to become an “agent” who does this homework problem (at least this is my original intention). I also gave it screenshots of the first three cells of the notebook to give it an idea of what to expect for the visualizations.

Notably, here, I had to spend significant effort to make sure the model doesn't ask for too many images. In other conversations, when I got to this point, the model asked for up to ~40 images in one prompt, which is not only not really feasible, but also can confuse the model as I've found. Therefore, I decided to limit the amount of images the model can ask for each time.

For the first sentence “The dog ran”:

1. Layer 0, Head 0 (Already provided)
2. Layer 5, Head 5
3. Layer 11, Head 11

For the second sentence “The dog sitting in the car”:

1. Layer 0, Head 0 (Already provided)
2. Layer 5, Head 5
3. Layer 11, Head 11

For the third sentence “Jaime Salazar is running for election. Despite his party affiliation, Mr.”:

1. Layer 0, Head 0 (Already provided)
2. Layer 5, Head 5
3. Layer 11, Head 11

Please generate these visualizations for me. This will help us observe how attention changes across different layers and heads for each sentence. Once I have these visualizations, I can start analyzing the patterns and differences in attention mechanisms.

Let’s proceed with generating these visualizations.

[USER]

Here are the requested results, with names being the sentence started underscore the requested image index (let me know if this was acceptable).

For the future, please give me a unique index for each image. Let’s also keep the amount of images each prompt to at most 6, so please think of more targeted visualizations from now.

Please review the images and analyze them. Let me know if there are further combinations you would like to analyze, or if you would like to answer questions 1 and 2.

[ASSISTANT]

Thank you for providing the visualizations. Let’s analyze the results and answer questions 1 and 2 based on the images provided.

Analysis of Visualizations:

Sentence 1: “The dog ran”

In my opinion, some reasonable choices for combinations of layer and head to visualize.

It does turn out that not all of the images were particularly useful, though one of the “dog sitting” images had an interesting attention from car to sitting.

Quite completely false. It seems like the model is quite confused by the fact that there are a lot of images given to it.

- **Layer 0, Head 0:** Each token attends to all other tokens, indicating a basic level of attention distribution.
- **Layer 5, Head 5:** The attention pattern is similar to Layer 0, suggesting that this head at this layer might be capturing more local context or similar features as the initial layer.
- **Layer 11, Head 11:** The attention pattern remains consistent, showing that even at the deepest layer, this head maintains a broad attention distribution.

Sentence 2: “The dog sitting in the car”

False and vague. The layer 5, head 5 image doesn’t show “more distributed attention” in any sense, as all tokens seem to primarily attend to “The”.

- **Layer 0, Head 0:** Each token attends to all other tokens, similar to the first sentence.
- **Layer 5, Head 5:** The attention is more distributed across the sentence, showing that this head at this layer might be capturing more complex relationships between tokens.
- **Layer 11, Head 11:** The attention pattern is similar to the previous layers, indicating that this head maintains a broad attention distribution across the sentence.

Sentence 3: “Jaime Salazar is running for election. Despite his party affiliation, Mr.”

The first one is false. The second one is somewhat false (tokens particularly attend to “Jai”). The third one is vague.

- **Layer 0, Head 0:** Each token attends to all other tokens, indicating a basic level of attention distribution.
- **Layer 5, Head 5:** The attention is more focused, particularly on the tokens “Jaime Salazar” and “election,” suggesting that this head at this layer might be capturing specific relationships or dependencies.
- **Layer 11, Head 11:** The attention pattern shows a more refined focus on certain tokens, indicating that this head at the deepest layer might be capturing the most complex dependencies or features.

Answers to Questions:

Earlier in my prompt, I told the model to ask me for more visualizations if it’s confused, but it didn’t, and instead just gave the answer.

1. **What similarities and differences do you notice in the visualizations between the examples in part (a)? Explore the queries, keys, and values to identify any interesting patterns associated with the attention mechanism.**
 - **Similarities:** Across all sentences and layers, each token attends to all other tokens, indicating a broad attention distribution. This suggests that the model captures both local and global context.
 - **Differences:** As we move from Layer 0 to Layer 11, the attention patterns become more refined and focused on specific tokens or pairs

of tokens. This indicates that deeper layers capture more complex dependencies and features. For instance, in the third sentence, deeper layers show a focus on “Jaime Salazar” and “election,” suggesting an understanding of named entities and their relevance.

2. How does attention differ between the different layers of the GPT model? Do you notice that the tokens are attending to different tokens as we go through the layers of the network?

- **Attention Differences:** The attention mechanism differs significantly between layers. In the initial layers (e.g., Layer 0), the attention is more distributed across all tokens, capturing basic context. As we move to deeper layers (e.g., Layer 5 and Layer 11), the attention becomes more focused on specific tokens or pairs of tokens, indicating the capture of more complex dependencies and features.
- **Token Attention:** Yes, tokens attend to different sets of tokens as we go through the layers. In deeper layers, certain tokens receive more focused attention, suggesting that the model is refining its understanding of the sentence structure and meaning.

This is the result for “one-shotting the problem” after prompt engineering effort.

Next Steps:

Based on this analysis, we can proceed to the next part of the notebook, which involves visualizing BERT’s attention mechanism. Please let me know which combinations of layers and heads you would like to visualize for the BERT model, and I will provide the corresponding visualizations.

[USER]

Before we move on, please be more careful about the statement that “each token attends to all other tokens”; is there a more precise way for you to describe the attention patterns? Can each token truly see every other token?

Further, consider the screenshot given here. Outside of all tokens attending to the “The” token, which other attention do you see here? Can you interpret that attention?

[ASSISTANT]

Apologies for the oversimplification in my previous responses. Let’s refine the analysis based on the visualizations provided.

Noting the fact that results are bad from letting the prior conversation, I strongly hinted the model closer to a correct solution. I also gave it a more targeted visualization example of “run” attending to “dog”.

Refined Analysis of Visualizations:

Sentence 1: “The dog ran”

- **Layer 0, Head 0:** The attention pattern shows that each token attends to itself and other tokens to varying degrees. Specifically, “The” attends to “dog” and “ran,” “dog” attends to “The” and “ran,” and “ran” attends to “The” and “dog.” This suggests that while there is some self-attention,

Still not correct; doesn’t capture the causal nature of the attention in GPT2.

It must have confused the additional example I gave with layer 5 head 5 of the original visualization example. The original layer 5 head 5 example didn't show the relationship between ran and dog.

there is also attention to other tokens, indicating the model is capturing relationships between words.

- **Layer 5, Head 5:** The attention pattern is similar to Layer 0, with each token attending to itself and other tokens. The attention distribution seems to be slightly more focused on the subject “dog” and the action “ran,” suggesting that this head might be capturing the subject-verb relationship.
- **Layer 11, Head 11:** The attention pattern remains consistent with the previous layers, indicating that this head maintains a broad attention distribution across the sentence.

Sentence 2: “The dog sitting in the car”

Still does not capture causality.

- **Layer 0, Head 0:** The attention pattern shows that each token attends to itself and other tokens. “The” attends to “dog,” “sitting,” “in,” “the,” and “car.” “dog” attends to “The,” “sitting,” “in,” “the,” and “car.” “sitting” attends to “dog,” “in,” “the,” and “car.” “in” attends to “dog,” “sitting,” “the,” and “car.” “the” attends to “dog,” “sitting,” “in,” and “car.” “car” attends to “dog,” “sitting,” “in,” and “the.” This suggests that the model is capturing the relationships between all tokens in the sentence.
- **Layer 5, Head 5:** The attention pattern is more focused, particularly on the tokens “dog,” “sitting,” “in,” and “car,” suggesting that this head at this layer might be capturing the phrase “dog sitting in the car” as a unit.
- **Layer 11, Head 11:** The attention pattern shows a refined focus on certain tokens, indicating that this head at the deepest layer might be capturing the most complex dependencies or features.

Sentence 3: “Jaime Salazar is running for election. Despite his party affiliation, Mr.”

Still does not capture causality.

- **Layer 0, Head 0:** The attention pattern shows that each token attends to itself and other tokens. The attention is distributed across the sentence, indicating a basic level of attention distribution.
- **Layer 5, Head 5:** The attention is more focused, particularly on the tokens “Jaime Salazar” and “election,” suggesting that this head at this layer might be capturing specific relationships or dependencies.
- **Layer 11, Head 11:** The attention pattern shows a more refined focus on certain tokens, indicating that this head at the deepest layer might be capturing the most complex dependencies or features.

Interpretation of Attention:

- The attention patterns indicate that the model is not simply attending to every other token equally. Instead, the attention is distributed in a

way that captures relationships between specific tokens. For example, in the sentence “The dog ran,” the subject “dog” and the verb “ran” have a stronger attention connection, indicating the model is capturing the subject-verb relationship.

- In the sentence “The dog sitting in the car,” the attention pattern suggests that the model is capturing the phrase “dog sitting in the car” as a unit, indicating an understanding of the phrase structure.

Next Steps:

Based on this refined analysis, we can proceed to the next part of the notebook, which involves visualizing BERT’s attention mechanism. Please let me know which combinations of layers and heads you would like to visualize for the BERT model, and I will provide the corresponding visualizations.

[USER]

Let’s change our plans. Let’s instead move question by question through the notebook, and for each question, I will provide you a small set of images that I find relevant to the question, and your task is to directly answer the question. Here is question 3: ” 3. Look at different layers of the BERT model in the visualizations of part (b) and identify different patterns associated with the attention mechanism. Explore the queries, keys, and values to further inform your answer. For instance, do you notice that any particular type of tokens are attended to at a given timestep? ”

I have provided two images for this. Please compare and contrast them, and then answer the question.

[ASSISTANT]

Based on the provided visualizations for BERT’s attention mechanism, let’s address question 3:

Question 3: Look at different layers of the BERT model in the visualizations of part (b) and identify different patterns associated with the attention mechanism. Explore the queries, keys, and values to further inform your answer. For instance, do you notice that any particular type of tokens are attended to at a given timestep?

Analysis: The two images provided represent attention visualizations from BERT for two different heads at Layer 2. The first image shows attention patterns for Head 10, while the second image shows attention patterns for Head 0.

1. Head 10 (Layer 2):

- The visualization is quite dense, indicating that each token attends to many other tokens across the sequence.

This part of analysis is correct, but this required me to find a good example and get rid of examples that don’t have much information.

I have given up on making the model completely autonomous to make its decisions as to what images it wants to visualize.

Here, I gave one image of many tokens attending to the start token and an example of most tokens attending to the next token.

Actually correct! This one is more dense than the other image.

Good!

- There is a strong attention pattern from the [CLS] token to all other tokens, which is typical as the [CLS] token is used for classification tasks and needs to gather information from the entire sequence.
- The attention distribution is relatively uniform across the sequence, suggesting that this head might be capturing a broad context or global features.

2. Head 0 (Layer 2):

Not really reflected in the image.

- The attention pattern is sparser compared to Head 10, with more distinct lines indicating specific attention relationships.
- Similar to Head 10, the [CLS] token attends to all other tokens, but the attention is more focused and less distributed.
- There are clear attention lines between tokens that are semantically related, such as “wanted” and “to,” “have” and “a,” and “bought” and “a cake.” This suggests that this head might be capturing more local or specific features, focusing on the grammatical structure or the relationships between adjacent tokens.

Conclusion:

- **Different Patterns:** The two heads show different attention patterns, with Head 10 having a more uniform and broad attention distribution, while Head 0 shows a more focused attention on specific, semantically related tokens.
- **Particular Types of Tokens:** At this timestep, the [CLS] token attends to all tokens, which is consistent across both heads. However, the attention patterns differ in how they distribute attention to other tokens. Head 0 seems to focus more on the grammatical structure and immediate relationships between tokens, while Head 10 captures a broader context.
- **Queries, Keys, and Values:** While the visualizations do not explicitly show queries, keys, and values, the attention patterns suggest that different heads are likely focusing on different aspects of the input sequence. Head 0 might be focusing on local features and immediate dependencies, while Head 10 might be capturing more global features or broader context.

This analysis shows that BERT’s multi-head attention mechanism allows different heads to focus on different aspects of the input, enabling the model to capture a variety of features and relationships within the data.

[USER]

To clarify, the token on the left is “what is attending”, and the token on the right is “what is being attended to”. Please look at the images more carefully again (which are provided again). One of it has most tokens attending to a

Here, I hint the model again.

certain other token, and the other has most tokens attending to one other token in a very patterned way. Can you identify which is which?

[ASSISTANT]

I apologize for the oversight. Let's analyze the two images more carefully based on the clarification provided:

Image Analysis:

1. Layer 2, Head 10:

- In this visualization, there is a dense network of attention connections from almost every token to the [CLS] token. This suggests that the [CLS] token is the focal point for gathering information across the sequence in this particular head.
- The [CLS] token attends to every other token, indicating its role in capturing the overall context of the sentence for classification purposes.

2. Layer 2, Head 0:

- Here, the attention connections are more structured and patterned. Most tokens attend to the token immediately following them in the sequence.
- For example, "wanted" attends to "to," "to" attends to "have," and so on. This pattern suggests that this head is capturing the sequential or positional relationships between tokens.

Correct analysis this time!

This took significant effort to get, though.

Identification:

- **Most tokens attending to a certain other token:** This pattern is observed in **Layer 2, Head 10**, where most tokens attend to the [CLS] token.
- **Most tokens attending to one other token in a very patterned way:** This pattern is observed in **Layer 2, Head 0**, where each token attends to the next token in a sequential manner.

Conclusion:

- **Layer 2, Head 10** focuses on the [CLS] token, which is typical for classification tasks where the [CLS] token is used to aggregate information from the entire input sequence.
- **Layer 2, Head 0** focuses on capturing the sequential order of tokens, which is crucial for understanding the grammatical structure and the flow of information within the sentence.

This detailed analysis should provide a clearer understanding of how different heads within the same layer of BERT can focus on different aspects of the input sequence, contributing to the model's ability to capture a wide range of linguistic features.

[USER]

This is a great observation for question 3! You're truly an expert at Transformers! Please continue being on fire.

Let's move on to question 4 now. Here it is: 4. Do you spot any differences between how attention works in GPT vs. BERT? Think about how the model architectures are different.

I have given you two more images for this; the first is a visualization from GPT2, and the second is a visualization from BERT. Please answer the question now.

[ASSISTANT]

Thank you for the kind words! Let's analyze the differences between how attention works in GPT-2 and BERT based on the provided visualizations and the inherent differences in their model architectures.

Question 4: Differences in Attention Mechanism between GPT-2 and BERT

GPT-2 Attention Visualization:

- **Layer 5, Head 5:** The visualization shows a pattern where each token attends to all other tokens in a relatively uniform manner. This is characteristic of the GPT-2 model, which uses a unidirectional transformer decoder. The attention mechanism in GPT-2 is designed to focus on the left context (previous tokens) when generating each token in the sequence. This is evident from the dense connections from each token to all previous tokens, facilitating the model's ability to generate text based on the context provided by earlier tokens in the sequence.

BERT Attention Visualization:

- **Layer 2, Head 0:** This visualization shows a more structured pattern of attention. Each token attends to itself and to subsequent tokens in the sequence. This reflects BERT's bidirectional nature, where the attention mechanism considers both the left and right context (previous and following tokens) when processing each token. This is crucial for BERT's ability to understand the context of a token within the entire sentence, which is essential for tasks like masked language modeling and next sentence prediction.

Architectural Differences:

Here, I encourage Kimi, as I've found that this helps for other LLMs to perform better. For this question, I gave an example of GPT2 all tokens attending to the first token, and an example of BERT with most tokens attending to the next token.

“uniformly to all other tokens” is ambiguous, though it does explicitly say “earlier tokens” later, which is good.

Good!

- **GPT-2:** Uses a unidirectional transformer decoder, which means that the attention mechanism only considers previous tokens in the sequence when generating each token. This is suitable for tasks like text generation where the model needs to predict the next token based on the context provided by earlier tokens.
- **BERT:** Utilizes a bidirectional transformer encoder, allowing the attention mechanism to consider both the left and right context when processing each token. This makes BERT well-suited for understanding the context of a token within the entire sentence, which is beneficial for tasks like sentiment analysis, question answering, and text classification.

Conclusion:

- **Attention Mechanism:** GPT-2's attention is unidirectional, focusing on the left context, while BERT's attention is bidirectional, considering both the left and right context.
- **Model Architecture:** GPT-2 is designed for text generation tasks, whereas BERT is designed for understanding the context within sentences, making it suitable for a wide range of NLP tasks that require understanding the meaning of text in context.

Correct! Though this may be partially due to Kimi already knowing what GPT2 and BERT are like.

These differences in attention mechanisms and model architectures lead to distinct capabilities and applications for GPT-2 and BERT in natural language processing tasks.

[USER]

Nice answer! Now let's move on to question 5.

Here is question 5: 5. Look through the different layers of the two BERT networks above associated with sentence a and sentence b, and take a look at the queries, keys, and values associated with the different tokens. Do you notice any differences in the embeddings learned for the two sentences that are essentially identical in structure but different in meaning?

The two sentences were "t was going to play at the park" and "i was going to play at the park". What token do you think behaves most differently between the two sentences based on the examples given?

Here, I gave two images of the two sentences, showing the keys and queries.

[ASSISTANT]

To answer question 5, let's analyze the provided visualizations from BERT for the two sentences:

1. "t was going to play at the park"
2. "i was going to play at the park"

Both sentences have a similar structure but differ in meaning due to the different subjects (“t” vs “i”). The visualizations show the attention patterns in Layer 10, Head 5 for both sentences.

Analysis:

Visualization 1: “t was going to play at the park”

- **Query (q):** The query vector for the token “play” shows a pattern of attention across the sentence.
- **Key (k):** The key vectors for each token in the sentence.
- **$q \times k$ (elementwise):** Element-wise multiplication of query and key vectors, highlighting the strength of attention between “play” and other tokens.
- **$q \cdot k$:** Dot product of query and key vectors, indicating the attention scores.
- **Softmax:** Normalized attention scores, showing the distribution of attention.

Visualization 2: “i was going to play at the park”

- Similar to the first visualization, but for the sentence with “i” as the subject.
- **Query (q):** The query vector for the token “play” shows a similar pattern of attention across the sentence.
- **Key (k):** The key vectors for each token in the sentence.
- **$q \times k$ (elementwise):** Element-wise multiplication of query and key vectors, highlighting the strength of attention between “play” and other tokens.
- **$q \cdot k$:** Dot product of query and key vectors, indicating the attention scores.
- **Softmax:** Normalized attention scores, showing the distribution of attention.

Differences in Embeddings:

- The token “play” behaves differently between the two sentences due to the different subjects. In the first sentence, “play” might be more associated with the context of “t” (which could be interpreted as “they” or “them”), while in the second sentence, “play” is associated with “i” (the speaker).
- The attention patterns and the resulting embeddings for “play” will reflect these differences in context and meaning. The model needs to adjust its

Very nice identification of the difference in how “play” works in the sentences, though I did also partially hint at this by giving it the dot product between play’s query and the keys of other tokens. Still, the very fact that it saw this means it could see/interpret what was going on.

Nice

understanding of “play” based on whether it refers to a third person or the speaker.

Conclusion:

- The token “play” shows different attention patterns and embeddings between the two sentences due to the different subjects. This demonstrates BERT’s ability to capture subtle differences in meaning based on context, even when the sentence structure is identical.

This analysis highlights how BERT’s attention mechanism and embeddings can vary based on the context provided by different subjects, allowing the model to understand and differentiate between similar sentences with different meanings.

[USER]

Okay, now let’s go to questions 6 and 7. They are: 6. Do you notice BERT’s bidirectionality in play? 7. Do you think pre-training the BERT helped it learn better representations?

One screenshot I think might be relevant is attached.

Here, I gave an image with many tokens attending to the next token, and some tokens attending to the initial <CLS> token.

[ASSISTANT]

Let’s address questions 6 and 7 based on the provided visualization and the inherent characteristics of BERT’s architecture.

Question 6: Do you notice BERT’s bidirectionality in play?

Answer: Yes, BERT’s bidirectionality is evident in the attention patterns displayed in the visualization. BERT’s architecture allows it to process information from both the left (previous tokens) and the right (subsequent tokens) context simultaneously. This is crucial for understanding the full context of a word within a sentence.

In the provided visualization:

- Each token attends to itself and other tokens, indicating that BERT is not only looking at the previous context (unidirectional, as in GPT-2) but also considering the following context.
- For instance, in the sentence “I bought a cake for it,” the token “bought” attends to “cake,” “for,” and “it,” showing that it’s considering both the action it’s related to (“cake”) and the purpose of the action (“for it”).
- This bidirectional processing enables BERT to capture a more comprehensive understanding of the sentence structure and semantics, which is essential for tasks like question answering and text classification where context from both directions is important.

This is not an accurate description of what’s shown in the visualization.

Question 7: Do you think pre-training the BERT helped it learn better representations?

Answer: Absolutely, pre-training BERT has significantly contributed to its ability to learn better representations of text data. Here's why:

- **Massive Data Exposure:** BERT is pre-trained on a large corpus of text data, allowing it to encounter a wide variety of linguistic patterns and structures. This extensive exposure helps it develop robust language representations.
- **Masked Language Modeling (MLM):** By predicting randomly masked words in sentences, BERT learns to understand the context in which words appear, improving its ability to represent the semantic meaning of words based on their surroundings.
- **Next Sentence Prediction (NSP):** This task helps BERT understand the relationship between sentences, enhancing its capability to represent the flow of information and the logical connections in text.
- **Transfer Learning:** Once pre-trained, BERT can be fine-tuned for specific tasks with a relatively small amount of task-specific data. This is because the pre-trained model already has a strong foundation in language understanding, needing only minor adjustments to excel in various NLP tasks.

The pre-training process, therefore, equips BERT with a deep understanding of language nuances, making it highly effective for a wide range of applications in natural language processing.

Conclusion:

The visualization you provided, along with BERT's architectural design, clearly demonstrates its bidirectional processing capability and the advantages gained from pre-training. These features make BERT a powerful tool for understanding and processing human language in various applications.

[USER]

Okay great. Now let's move on to questions 8 and 9; here they are: 8. Do you notice different features being learned throughout the different attention heads of BERT? Why do you think this might be? 9. Can you identify any of the different features that the different attention heads are focusing on?

The notebook visual basically visualized the attention across all different layers (vertical axis) at all different heads (horizontal axis). I have attached an "overview" picture as well as a "specific view" picture to this chat. Please answer those two questions after CAREFUL analysis.

[ASSISTANT]

I find this answer to be acceptable.

Here, I gave it a larger screenshot containing a large amount of the attention visualizations at different heads/layers, and also two of them side-by-side that are clearly different.

Based on the visualizations provided, we can analyze the attention patterns across different layers and heads of BERT to answer questions 8 and 9.

Question 8: Do you notice different features being learned throughout the different attention heads of BERT? Why do you think this might be?

Answer: Yes, different attention heads in BERT appear to learn different features. This is evident from the diverse attention patterns observed across various heads and layers in the visualizations.

- **Diversity in Attention Patterns:** The visualizations show a wide variety of attention patterns across different heads. Some heads show dense, uniform attention across the sequence, while others display more sparse, focused attention. This diversity suggests that each head is specializing in capturing different aspects of the input data.
- **Specialization of Attention Heads:** The multi-head attention mechanism in BERT allows each head to focus on different types of relationships or features within the data. This specialization enables the model to jointly attend to information from different representation subspaces at different positions, which is a key aspect of the Transformer architecture.
- **Reason for Specialization:** The reason behind this specialisation is to enable the model to capture a rich variety of features and relationships within the data. By allowing each head to focus on different aspects, BERT can learn a more comprehensive and nuanced representation of the input text.

Good answer.

Question 9: Can you identify any of the different features that the different attention heads are focusing on?

Answer: While it's challenging to directly interpret the specific features each attention head is focusing on without further analysis, we can make some educated guesses based on the attention patterns observed:

- **Local Features:** Some heads show attention patterns that are more focused on adjacent tokens, suggesting they might be capturing local features such as grammar, syntax, or immediate contextual relationships.
- **Global Features:** Other heads exhibit more distributed attention across the entire sequence, indicating they might be capturing global features like overall sentence structure, topic, or long-range dependencies.
- **Semantic Relationships:** Certain heads might be specializing in identifying semantic relationships between words, such as subject-verb agreement, or the connections between entities mentioned in the text.
- **Syntactic Dependencies:** Some attention heads could be focused on capturing syntactic dependencies, such as the relationships between nouns

Good answer as well; actually does capture the contrast between the consecutive attention visualizations.

and their modifiers, or verbs and their objects.

- **Positional Information:** Given BERT's use of positional embeddings, some heads might be attending to the positional information of tokens to understand their order within the sequence.

Conclusion:

The visualizations provide evidence that different attention heads in BERT learn to focus on diverse features and aspects of the input data. This specialization allows BERT to develop a rich and multifaceted representation of the text, capturing various linguistic phenomena and relationships. While it's difficult to pinpoint the exact features each head is focusing on without further analysis, the attention patterns suggest a wide range of linguistic and contextual information being captured by the different heads.

[USER]

Makes sense! The rest of the questions are about completely initializing BERT and seeing attention. Now let's go to question 10, which is the following: 10. What differences do you notice in the attention patterns between the randomly initialized and trained BERT models?

I have re-initialized the model and generated an attention pattern, which is attached. Please answer this question.

[ASSISTANT]

To address question 10, let's compare the attention pattern from a randomly initialized BERT model with the attention patterns we've observed from the trained BERT model.

Question 10: What differences do you notice in the attention patterns between the randomly initialized and trained BERT models?

Answer: The provided visualization from the randomly initialized BERT model shows a distinct pattern compared to the trained model:

- **Randomly Initialized Model (Attached Visualization):**
 - The attention pattern appears more uniform and less structured. This is because the weights are randomly initialized, and the model has not yet learned to focus on specific linguistic features or relationships within the text.
 - The attention connections are more evenly distributed across the sequence, without clear focal points or patterns that indicate an understanding of the sentence structure or semantics.
 - This uniform distribution suggests that the model has not yet developed the ability to distinguish between important and less important

relationships between tokens.

- **Trained Model:**

- In contrast, the visualizations from the trained BERT model displayed more structured and focused attention patterns.
- The trained model shows clear attention connections that reflect an understanding of the sentence's grammatical structure and semantic relationships. For example, in the visualizations provided earlier, we observed attention focusing on specific tokens that are related to each other in meaningful ways (e.g., subject-verb relationships, object of prepositions).
- The trained model's attention patterns indicate that it has learned to capture both local (adjacent tokens) and global (long-range dependencies) features within the text.

Conclusion:

The key differences between the randomly initialized and trained BERT models are evident in the attention patterns:

- The **trained model** exhibits attention patterns that reflect an understanding of language structure and semantics, with focused attention on meaningful relationships between tokens.
- The **randomly initialized model** shows a more uniform and less informative attention pattern, lacking the structured focus that comes from learning during training.

This analysis highlights the importance of pre-training in developing BERT's ability to effectively process and understand natural language. The attention mechanism in a trained BERT model is a powerful tool for capturing complex linguistic features, which is crucial for its high performance across a wide range of NLP tasks.

[USER]

Makes sense. Let's now move on to our last and final question: question 11, which is this: 11. Run the final cell in the notebook. What are some words or tokens that you would expect strong attention between? What might you guess about the gradients of this attention head for those words?

The two sentences passed in as a string to BERT are "I am happy" and "I am sad". I have attached a screenshot of the current (untrained) attention for it, as well as a prior trained attention example.

Here, I just gave it a visualization of untrained BERT (uniform) and a visualizaion of trained BERT (very specific attention).

[ASSISTANT]

To address question 11, let's analyze the attention patterns for the sentences "I am happy" and "I am sad" in both the trained and untrained BERT models.

Question 11: What are some words or tokens that you would expect strong attention between? What might you guess about the gradients of this attention head for those words?

Answer: Based on the provided visualizations and the nature of the sentences, here's what we can infer:

- **Expected Strong Attention:**

- In both sentences, we would expect strong attention between the subject “I” and the predicate “happy” or “sad.” This is because the relationship between the subject and its emotional state is crucial for understanding the meaning of the sentences.
- Additionally, there might be strong attention between “am” and the following adjectives “happy” or “sad” due to the grammatical structure of the sentences.

- **Gradients of Attention:**

- In the **trained model**, the gradients of attention for these key relationships (“I” to “happy”/“sad,” “am” to “happy”/“sad”) would likely be steep and well-defined. This is because the model has learned to associate these tokens strongly based on the large amount of data it was trained on. The attention mechanism would have been optimized to capture these important relationships, leading to high attention weights for these connections.
- In the **untrained model**, the gradients of attention might be more uniform or less distinct. Since the model has not yet learned from data, it does not have the capacity to differentiate between important and less important relationships as effectively. The attention distribution might be more even across tokens, without the sharp peaks that indicate strong associations in a trained model.

Conclusion:

- The visualizations suggest that in a trained BERT model, we would expect to see strong attention between the subject “I” and the emotional adjectives “happy” or “sad,” as well as between “am” and these adjectives. These relationships are fundamental to capturing the meaning of the sentences.
- The gradients of attention for these key relationships in a trained model would likely be steep, indicating strong learned associations. In contrast, an untrained model would show a more uniform attention distribution, with less pronounced gradients, reflecting its lack of learned relationships.

This analysis underscores the impact of training on the attention mechanism of BERT, highlighting how pre-training enables the model to develop focused and

Reasonable analysis for
this question.

meaningful attention patterns that are crucial for language understanding.

[USER]

Okay this makes sense! Now please summarize your answers for the questions into 1 paragraph each. Please give your answers as one paragraph for each question now; questions are here: 1. What similarities and differences do you notice in the visualizations between the examples in part (a)? Explore the queries, keys, and values to identify any interesting patterns associated with the attention mechanism. 2. How does attention differ between the different layers of the GPT model? Do you notice that the tokens are attending to different tokens as we go through the layers of the network? 3. Look at different layers of the BERT model in the visualizations of part (b) and identify different patterns associated with the attention mechanism. Explore the queries, keys, and values to further inform your answer. For instance, do you notice that any particular type of tokens are attended to at a given timestep? 4. Do you spot any differences between how attention works in GPT vs. BERT? Think about how the model architectures are different. 5. Look through the different layers of the two BERT networks above associated with sentence a and sentence b, and take a look at the queries, keys, and values associated with the different tokens. Do you notice any differences in the embeddings learned for the two sentences that are essentially identical in structure but different in meaning? 6. Do you notice BERT's bidirectionality in play? 7. Do you think pre-training the BERT helped it learn better representations? 8. Do you notice different features being learned throughout the different attention heads of BERT? Why do you think this might be? 9. Can you identify any of the different features that the different attention heads are focusing on? 10. What differences do you notice in the attention patterns between the randomly initialized and trained BERT models? 11. Run the final cell in the notebook. What are some words or tokens that you would expect strong attention between? What might you guess about the gradients of this attention head for those words?

Questions that I asked you together can be answered together in one paragraph (e.g. questions 6 and 7, and questions 8 and 9).

[ASSISTANT]

1. **Similarities and Differences in Visualizations (Part a):** The visualizations of attention patterns in part (a) reveal both similarities and differences. Similarities include the presence of attention connections between all tokens, indicating that each token considers the context provided by others. Differences are observed in the distribution and focus of attention, where certain tokens, like the subject or verb, receive more attention, suggesting the model captures subject-verb relationships. This pattern indicates that the attention mechanism is sensitive to the grammatical structures and semantic roles of tokens within the sentence.
2. **Attention Differences Across GPT Layers:** The attention in GPT-2 differs across layers, with initial layers capturing more local context

I ask the model to summarize its conclusions for each of the questions.

Still not solid about the uni-directionality part.

Some general answer that doesn't seem too influenced by the images.

The rest seem relatively good.

and deeper layers focusing on broader context. This shift reflects the hierarchical nature of the transformer architecture, where earlier layers learn to recognize local patterns and later layers integrate these into a more comprehensive understanding of the sequence.

3. **Patterns in BERT Attention (Part b):** BERT's attention mechanism exhibits diverse patterns across different layers. In part (b), we see that attention connections are not only between adjacent tokens but also skip over others, indicating the model's ability to capture both local and long-range dependencies. This bidirectional attention allows BERT to understand the context of a token from both directions, enhancing its ability to grasp sentence meaning.
4. **Attention Differences Between GPT and BERT:** GPT-2 and BERT have distinct attention mechanisms due to their architectural differences. GPT-2's unidirectional attention is suited for text generation, focusing on the left context. In contrast, BERT's bidirectional attention enables it to consider the full context around each token, which is beneficial for tasks requiring comprehensive understanding of sentence structure and meaning.
5. **Embeddings for Similar Sentences in BERT:** Despite the structural similarity, the embeddings for sentences "t was going to play at the park" and "i was going to a play in the park" differ in BERT due to the different subjects. The model adjusts its attention based on the context provided by the subject, leading to distinct embeddings that reflect the unique meaning of each sentence.
- 6-7. **BERT's Bidirectionality and Pre-training Benefits:** BERT's bidirectional attention mechanism allows it to process information from both left and right contexts, which is evident in the attention patterns observed. This capability, combined with its pre-training on large datasets, equips BERT with robust representations that capture nuanced language features and relationships, enhancing its performance across various NLP tasks.
- 8-9. **Feature Learning Across Attention Heads in BERT:** Different attention heads in BERT learn to focus on different features, which could range from local grammatical structures to long-range semantic relationships. This specialization allows BERT to develop a rich and multifaceted representation of the input text, capturing a wide array of linguistic phenomena.
10. **Attention Patterns in Randomly Initialized vs. Trained BERT:** The attention patterns in a randomly initialized BERT model are more uniform and lack the structured focus seen in the trained model. The trained model exhibits attention patterns that reflect an understanding of language structure and semantics, with clear focal points indicating the model's ability to capture meaningful relationships between tokens.
11. **Expected Attention Between Words/Tokens:** In the final cell of

the notebook, for the sentences “I am happy” and “I am sad,” we would expect strong attention between the subject “I” and the predicate adjectives “happy” or “sad,” due to their semantic relationship. Additionally, the gradients of this attention head for these words would likely be steep in a trained model, indicating a strong learned association, while in an untrained model, the gradients might be more uniform, reflecting the lack of learned relationships.